The Cohomology of Equivocation

What Cycle Residuals Certify in Federated Witness-Log Gossip

Paper 7 of the Harbor program — observability, counterexamples, and the consistency radius

Erich Owens
The Harbor Research Program erich@curiositech.ai

August 2026

Abstract

A cycle residual is a sound test of whether observed edge differences admit one vertex assignment, but its diagnostic value depends on precisely what evidence the analyst holds. For prefix-coordinate edge differences on a visible graph K, $r = \operatorname{dist}(g_K, \operatorname{im}\delta_K)$ is the norm of the cycle-space projection. A single nonzero fault on edge e has $r = |s|\sqrt{1 - R^{K_c}_{\text{eff}}(e)}$: it is detectable exactly when e lies on a cycle of the visible coordinate graph K_c. This criterion does *not* extend to arbitrary split views by one witness; a triangle with a leaf supplies an explicit counterexample. For sender q, the number of visible-neighbor components after deleting q, minus one, is the exact dimension of invisible split views in each coordinate. Under a full projected error ball of radius ϵ, a one-edge fault is distinguishable from honest data exactly when $|s|\sqrt{1 - R^{K_c}_{\text{eff}}(e)} > 2\epsilon$; a sender-wide criterion combines the blind-spot dimension with a least singular value. For an additional edge-only receipt, we give an exact marginal residual formula: a bridge into another visible component cannot raise the score immediately, while a same-component receipt raises its squared score by its squared least-squares innovation divided by one plus the endpoints' effective resistance. More decisively, if the analyst receives both authenticated endpoint reports on each relayed link, it can compare a sender's reports directly. Every nonzero cycle residual then implies an inconsistency that this equal-information checker can already detect. The synthetic harness's "cohomology-only" counts compare against a weaker checked-edge baseline and do not establish a detection advantage. The residual remains useful as a quantitative, topology-aware consistency score when only authenticated edge-difference receipts are retained, but an exact cycle-sum test has the same noiseless detection set. We give the exact information boundary, the effective-resistance sensitivity law, and a benchmark contract needed to test whether scoring, localization, or audit cost yields operational value. No deployed signed-receipt system is evaluated here.

1 What the analyst can know

A federation sends authenticated witness-log reports along a graph. An auditor wants to know whether one witness gave incompatible reports to different peers. The evidence question asks which authenticated statements reach the auditor. The algebraic question asks whether derived edge differences fit one assignment of states to vertices. A cycle residual answers the second question; the first determines whether that answer adds information.

Suppose witness q sends one log-head claim to a and a different claim to b. If both signed claims reach the auditor with the same source, coordinate, and snapshot identifier, a direct duplicate-claim check detects the conflict. The check does not require a and b to compare with each other. A detector that discards those claims and retains only edge differences cannot fairly claim an advantage over the direct check.

A constrained setting still merits study. An auditor may retain authenticated edge-difference receipts while raw endpoint claims are unavailable. It can then test whether visible cycle sums

vanish. The residual gives a magnitude and a sensitivity map; topology states where the test is blind. The exact evidence contract is part of the result.

2 Cochains, sections, and cycles

A cellular sheaf assigns a vector space to each vertex and edge and a restriction map from each incident vertex into the edge space. The paper uses coordinate-selection maps. Its real-valued stalks model *features* of log reports, conditional on an equality-preserving map from authenticated claims; they are not raw cryptographic log heads.

The coboundary δ maps a vertex assignment x to its oriented edge differences. For one scalar coordinate, δ is the signed incidence matrix. The assignment x is a *zero-cochain*; the pattern δx is a *coboundary*. A *global section* of the unshifted sheaf instead satisfies $\delta x = 0$. The detector asks whether the observed edge cochain g lies in $\operatorname{im}\delta$, meaning that some vertex assignment explains its differences. It does not assert that the explaining assignment is a global section of the unshifted sheaf, or that the sender claims were mutually consistent.

A signed sum of a coboundary around a graph cycle telescopes to zero. The quotient $H^1 = \operatorname{coker}\delta$ describes edge data modulo coboundaries; in the coordinate-selection graph model, its Euclidean representative is the projection onto the cycle space. The existence of nonzero H^1 says the topology admits an obstruction. It does not say the observed execution has one: only the class of g can do that.

The completion residual r is the Euclidean distance from g to $\operatorname{im}\delta$ after severed edge rows are omitted. Effective resistance of an edge is the voltage drop across it under one unit of injected current in the unit-resistor graph. It governs how strongly a *single-edge* perturbation survives this projection. These objects can be computed with incidence matrices and least squares; the cohomological language identifies their structural meaning without adding information to the observed receipts.

3 The model and the observability contract

The sheaf. Fix a finite gossip graph $G = (V, E)$. Witness v carries a local log state $x_v \in \mathbb{R}^{d_v}$ (its stalk). Each edge $e = \{u, v\}$ carries the *shared prefix* the two endpoints both report — a subset S_e of coordinates, $|S_e| \le L$ — and the restriction maps $\rho_{v,e} : \mathbb{R}^{d_v} \to \mathbb{R}^{|S_e|}$ are coordinate-subset selections. The observable on edge e is the disagreement $g_e = \rho_{u,e}(u\text{'s report}) - \rho_{v,e}(v\text{'s report})$. A signed head requires an explicit equality-preserving feature map before this real-valued subtraction: equal heads map equally, and a head inequality claimed detectable in coordinate c must map to an inequality in c. No such signed-receipt implementation is evaluated here. Receipts must bind source, peer, coordinate, snapshot, and provenance before statements can be compared. Honesty means all sender reports are consistent with one vertex assignment x: then $g = \delta x$ is a coboundary of the sheaf coboundary δ, and every cycle-sum of g vanishes. Equivocation is an offset pattern ε added to q's per-neighbor reports; the question for the derived edge data is what survives in H^1 — the obstruction space $\operatorname{coker}(\delta)$, which for prefix restrictions decomposes per shared coordinate c into the cycle spaces of the *coordinate subgraphs*

$$G_c \;=\; \big(V,\ \{e \in E :\ c \in S_e\}\big)$$

[verified framework, [3, 4]]. G_c is what the *federation* has: every link that carries coordinate c. What the *edge-only analyst* has is smaller when links are severed; see K_c below.

Three-tier visibility — the observability contract. Not every edge is equally visible to the analyst, and the paper's claims are stated *per tier* (Figure 1):

- **compared** — the analyst holds the direct cross-check of the edge: pairwise detection, no cohomology needed;

- **relayed** — both authenticated endpoint claims reach the analyst by gossip, but the endpoints did not cross-check. Their arrival permits both a direct duplicate-claim check by sender and a derived edge difference;
- **severed** — no report about the edge crosses to the analyst at all: the corresponding data block is a free variable of the edge-data completion. Other authenticated claims from that sender may still be available elsewhere.

The visible coordinate subgraph K_c. Write K for the compared-plus-relayed edge set — every link the analyst holds a report about — and

$$K_c \;=\; \big(V,\ \{e \in K :\ c \in S_e\}\big) \;\subseteq\; G_c$$

for its coordinate-c part: the coordinate subgraph with the severed links deleted. Every *one-edge residual* criterion is a statement about cycles of K_c, never of G_c, and the reason is one line of algebra made explicit in the detector below — a severed block is a free variable, so minimising over it simply deletes its row, and a loop that runs through a severed link is not a loop the analyst can close. The federation's topology sets G_c; the observability contract sets K_c; only the second enters this edge-data residual.

The detector. The *completion residual* is the least-squares defect of the best global explanation of everything visible:

$$r \;=\; \min_{x,\ g_{\mathrm{sev}}} \big\|\, g_K - (\delta x)\big|_K \,\big\|_2,$$

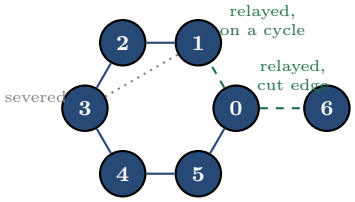
the minimum over vertex assignments x and free severed blocks g_{sev}, where K is the compared-plus-relayed edge set. The severed blocks are wholly unconstrained, so minimising over them is exactly *dropping their rows*: g_{sev} never enters the objective, and $r = \mathrm{dist}\big(g_K,\ \mathrm{im}\,\delta_K\big)$ for δ_K the coboundary restricted to visible edges. Honest executions give $r = 0$ exactly, for every visibility pattern; $r > 0$ certifies that *no* global assignment explains the visible edge differences. Its size depends on the feature map, coordinate scaling, and metric; it is not a cryptographic measure of log divergence. Equivalently $r = \|\Pi_K\, g_K\|$, the projection of the visible disagreement data onto the per-coordinate cycle spaces *of the visible subgraphs*:

$$r^2 = \sum_c \big\|\mathrm{Proj}_{\mathrm{cycle}(K_c)}\, g^c\big\|^2, \qquad g^c \in \mathbb{R}^{K_c} \text{ the coordinate-}c \text{ part of the visible data}$$

[internal, `sheaf_harness_v2.py`]. The derivation is three lines and is the whole bridge from the minimization to Theorem 1, so it is worth stating: prefix restrictions are coordinate selections, so δ_K block-diagonalises per coordinate into the signed incidence operator B_c of K_c — the severed rows having already dropped out; edge space splits orthogonally as $\mathbb{R}^{K_c} = \mathrm{im}(B_c) \oplus \ker(B_c^\top)$, cut space plus cycle space; and the residual is what is left after removing everything an honest world could have produced, i.e. the component in $\mathrm{cycle}(K_c)$. Writing G_c here instead would not merely be loose, it would fabricate detections: on C_6 with a lie of 3 on a relayed edge and one other cycle edge severed, the residual is 1.1×10^{-14} while the G_c form reports 1.2247 — and reports a different value again for a different honest world, so it is not even a well-defined statistic [internal, `sheaf_consistency_radius.py`].

4 The mechanism: visible cycles and their exact scope

A one-edge fault is visible to the residual exactly when its edge lies on a visible coordinate cycle. A general split view by one sender obeys a different criterion, and authenticated relayed claims admit direct comparison.



one gossip graph, three visibility tiers

K_c = compared ∪ relayed

what each tier supplies to the analyst

—— compared	analyst holds a direct endpoint cross-check for the edge and can inspect its authenticated claims.
- - - relayed	both endpoint claims reach the analyst without an endpoint cross-check. A direct sender-claim check uses them. For a *single-edge* fault, the residual fires on a visible cycle and vanishes on a bridge.
····· severed	no edge report reaches the analyst; its row drops out of K_c. A fault confined to that row cannot affect the edge-data residual.

Figure 1: The three-tier observability contract — the tiers are properties of the *analyst's evidence*, not of the federation, and all three coexist on one topology. Left: one gossip graph wearing all three tiers — harborblue solid compared, seagreen dashed relayed, gray dotted severed; the compared-plus-relayed links are the visible subgraph K_c that the one-edge residual criterion uses. The severed link 1–3 is drawn as a chord. Right: the evidence supplied by each tier. The relayed tier already supplies both claims to an equal-information checker. The visible-cycle criterion describes the residual for one edge fault; it does not cover every one-sender split view. Figure 2 uses this same legend [internal, `sheaf_harness_v2.py`].

Theorem 1 (one-edge criterion). Fix a coordinate c, a visible edge e of K_c, and a nonzero edge perturbation s. With unit edge weights and the Euclidean metric,

$$r = |s|\sqrt{1 - R_{\text{eff}}^{K_c}(e)}.$$

Consequently $r > 0$ exactly when e lies on a cycle of K_c. A bridge has effective resistance 1 and zero residual. A severed edge contributes no observed row. This is a theorem about a *one-edge* perturbation, not every split view by a single sender.

General one-sender criterion. Let $A_q o$ place sender q's per-neighbor coordinate offsets on visible incident edges, with orientation signs. Then $r = \|\Pi_K A_q o\|$, so detection is exactly $\Pi_K A_q o \neq 0$. For each coordinate, $\ker(\Pi_K A_q)$ comprises offset patterns constant on edges from q into each connected component of $K_c - q$. Constants for distinct components may differ. Indeed, $\Pi_K A_q o = 0$ exactly when $A_q o = B_c x$ for a potential x. Every nonincident edge has zero offset, forcing x to be constant on each component of $K_c - q$; those constants also construct the converse potential.

Blind-spot dimension. Let d_q^c be the number of visible c-edges at q, and let h_q^c be the number of connected components of $K_c - q$ touched by those edges. If $d_q^c > 0$, then

$$\dim \ker(\Pi_K A_q)^c = h_q^c, \qquad b_q^c := \dim\big(\ker(\Pi_K A_q)^c / \langle \mathbf{1} \rangle\big) = h_q^c - 1.$$

Set $b_q^c = 0$ when $d_q^c = 0$. The quotient removes a uniform sender-state change, leaving precisely the independent *invisible split-view* directions. Thus every nonuniform one-sender offset in coordinate c is detected exactly when $b_q^c = 0$. The total blind-spot dimension for independent coordinates is $B_q = \sum_c b_q^c$. Under an edge-only receipt design that may add c-bearing edges among the touched components of $K_c - q$, at least b_q^c such edges are needed to remove this blindness, and b_q^c suffice if arbitrary cross-component links are admissible. Each added edge can merge at most two components; a spanning tree on the components attains

the bound. This counts observability, not authenticated-claim consistency or Byzantine fault tolerance.

Counterexample to a universal split-view criterion. Let q, a, b form a triangle and attach leaf l to q, all edges visible in one coordinate. Starting from zero states, q sends offset 1 to a, b and 0 to l. The alternative potential $x_q = 1$, $x_a = x_b = 0$, $x_l = 1$ produces exactly these edge differences: the two triangle edges at q differ by 1, while $a - b$ and $q - l$ differ by 0. Hence $r = 0$ despite a split view and a visible cycle through q. The components of $K_c - q$ are $\{a, b\}$ and $\{l\}$, so this is precisely a nonuniform kernel offset. A duplicate-claim checker with both authenticated claims from q detects it directly if they bind the same coordinate and snapshot.

The cochain by hand. On a visible four-relay ring in one scalar coordinate, take the oriented edge differences $g = (0, 0, 0, 3)$. A coboundary sums to zero around the ring; g sums to 3. The nearest potential $x = (0, 0.75, 1.5, 2.25)$ gives $\delta x = (-0.75, -0.75, -0.75, 2.25)$, leaving magnitude 0.75 on each edge. Thus

$$r = \sqrt{4(0.75)^2} = 1.5 = 3/\sqrt{4} = 3\sqrt{1 - R_{\text{eff}}^{C_4}(e)}, \quad R_{\text{eff}}^{C_4}(e) = 3/4.$$

On a path, or on the ring with any other edge severed, the same one-edge fault is a coboundary and $r = 0$. On a fully visible C_6, a fault of size 3 gives $3/\sqrt{6} = 1.2247$; severing another edge makes the residual zero. These exact scalar examples preserve the valid effective-resistance mechanism.

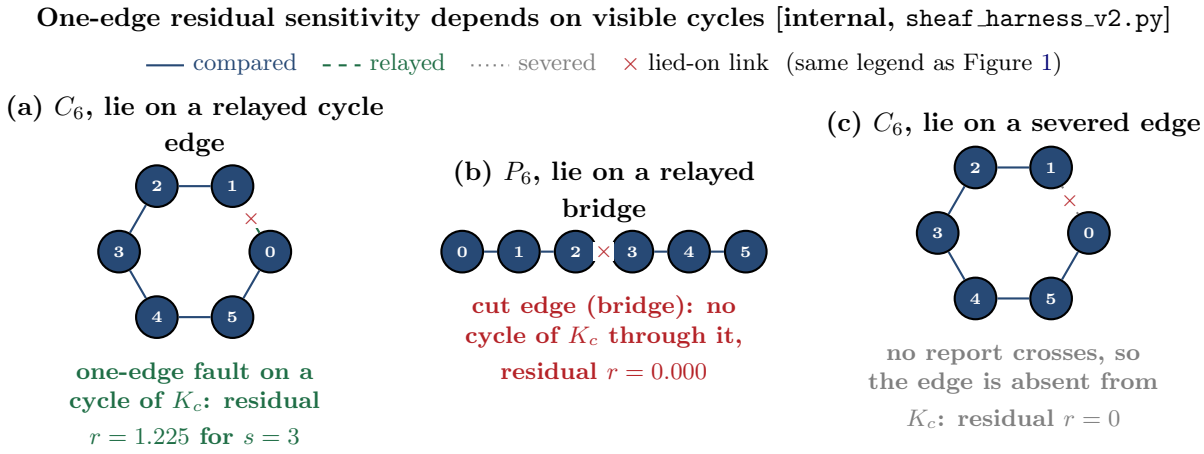


Figure 2: Regime diagram for the mechanism, in one legend with Figure 1 (harborblue solid compared, seagreen dashed relayed, gray dotted severed; the shipred × marks the lied-on link, independently of its tier). (a) A single-edge fault on a relayed link in a visible cycle gives $r = 1.225$ for $s = 3$. (b) The same fault on a relayed bridge gives $r = 0$. (c) A severed edge has no row in K_c and cannot affect this residual. The relayed endpoint claims in (a) and (b) also permit direct sender-claim comparison when authenticated and snapshot-aligned. The figure states a one-edge residual criterion, not a unique detection advantage or a criterion for every split view [internal, `sheaf_harness_v2.py`].

Equal-information baseline. In the stated relayed tier the analyst holds authenticated endpoint reports. It can group a sender's claims by coordinate and snapshot and test equality without asking the endpoints to compare. If every sender made one consistent claim, all derived edge differences would be a coboundary; therefore any nonzero cycle residual implies an inconsistency that this direct checker can also detect. An edge-only receipt pipeline may lose the raw claims; there the residual still gives a metric score and residual pattern. With the same edge-only data, exact oriented cycle sums have the identical noiseless zero/nonzero detection set. The residual does not create additional detection information.

Proposition 2 (equal-information dominance). Assume each visible edge record contains both authenticated endpoint claims for the same coordinate and snapshot, and the edge cochain is their difference. If a sender's claims agree across all its visible incident edges for each coordinate, then $g_K = \delta_K x$ for a vertex assignment x and $r = 0$. Consequently a nonzero residual implies a conflict that direct duplicate-claim checking on the *same records* detects. The converse fails, including in the triangle-with-leaf example.

Proof. Define x_v^c as sender v's common claim for coordinate c; choose any value when v has no visible incident edge carrying c. Every visible edge difference is then $x_u^c - x_v^c$, so $g^c = B_c x^c$ and its cycle projection vanishes. Contraposition proves the detection implication.

A zero residual means the visible edge differences admit a potential, not that the actual reports came from one head. A nonzero residual proves inconsistency of the encoded, snapshot-bound data, not who caused it. Both interpretations require authenticated provenance and a feature map that preserves the head equality being tested.

5 The harness: what its trials establish

The synthetic harness uses 200 trials per arm at seed 20260816, real five-dimensional stalks, and shared coordinate subsets. The checked-edge baseline tests whether a directly checked edge exposed the fault. A trial labeled cohomology-only means the residual was positive while this restricted baseline was silent. The harness also reconstructs each sender's visible coordinate claims and runs Proposition 2's equal-information duplicate-claim check.

Observed behavior. In the two-path split-view arm the residual was positive and the checked-edge baseline silent in 200/200 trials. In the expander arm the corresponding count was 113/200, with 87 residual-dark trials. Of those 87, 79 had no relayed incident fault edge and 8 had a relayed fault edge on no visible coordinate cycle. The severed-cycle arm gave 0/200; a cut-edge arm had maximum numerical residual 1.5×10^{-13} over 400 trials. The no-equivocator control gave 0/200. Across all nine arms and 1,800 trials, the equal-information checker found *zero* residual-only cases and *209* direct-only cases. These counts support Proposition 2 within the synthetic feature and snapshot model; they are not a deployed signed-log evaluation.

The counts depend on the sampled topology, offsets, metric, and threshold. The harness checks the blind-spot dimension against independent sender-map nullity calculations in all 224 four-vertex graph/sender cases with at least one incident edge. The equal-information comparison has been run on these synthetic records; exact cycle sums and noisy scoring have not been compared. An operational benchmark must predefine authenticated receipt and snapshot binding, the equality-preserving feature map, noise, false-alarm threshold, runtime cost, and localization target, then run all three evaluators on matched evidence.

6 The consistency radius: certificate, sensitivity, and cost

The residual is the least Euclidean unrestricted edge perturbation needed to reconcile visible data. A perturbation constrained to one sender can require more, or be infeasible. Effective resistance supplies exact one-edge sensitivity.

CR-1 (soundness and constrained lower bound). Let Π_K project onto $\ker(\delta_K^\top)$ in

the chosen Euclidean metric. Then

$$r = \min_x \|g_K - \delta_K x\| = \|\Pi_K g_K\|.$$

Thus $r > 0$ exactly when no potential explains the visible differences, and honest data give $r = 0$. The minimum norm of an *unrestricted edge-data perturbation* making g_K a coboundary is exactly r, achieved by $-\Pi_K g_K$. If the perturbation must originate at a fixed witness q, the feasible minimum is

$$m_q(g) = \inf\{\|A_q o\| : \Pi_K A_q o = \Pi_K g_K\} \geq r,$$

with $m_q(g) = +\infty$ when the constraint is infeasible. The minimum norm of the offset vector o is a separate quantity. With $o_\perp$ orthogonal to $\ker(\Pi_K A_q)$,

$$\sigma^+_{\min}(\Pi_K A_q)\|o_\perp\| \leq \|\Pi_K A_q o\| \leq \|\Pi_K A_q\|\|o_\perp\|.$$

Kernel offsets can be arbitrarily large while r remains zero.

CR-2 (one-edge sensitivity). For a visible unit-weight edge e in coordinate c,

$$\|\Pi_K(s\mathbf{e}_e)\| = |s|\sqrt{1 - R^{K_c}_{\text{eff}}(e)}.$$

Its projected support lies on visible cycles containing e. For general sender offsets, support lies in the union of cycle-containing blocks reached by incident offsets; cancellation can remove some or all signal, and the pattern cannot attribute a sender. Additive edge noise η changes the residual by at most $\|\Pi_K \eta\|$. A threshold requires a justified noise bound or calibration.

CR-3 (coordinate decomposition). Let $C = \bigcup_{e \in K} S_e$ and $T = |C|$. Coordinate selection splits the projection into at most T scalar graph-Laplacian systems, one for each nonempty K_c. The bound is at most L only if all edge coordinate sets draw from one globally common set of at most L coordinates. Cost depends on $\sum_c |K_c|$, conditioning, tolerance, and solver. Near-linear sparse-solver theory [11] is relevant but does not prove a per-round runtime for this federation.

Numbers by hand. On C_n, the edge is in parallel with a path of $n-1$ unit resistors, so $R_{\text{eff}}(e) = (n-1)/n$ and a one-edge size-s perturbation gives $r = |s|/\sqrt{n}$. On C_6, the projected norm per unit one-edge fault is $1/\sqrt{6} \approx 0.4082$ and $s = 3$ yields 1.2247; on C_{12}, $s = 0.5$ yields about 0.1443. This 0.4082 is a one-edge sensitivity, not a universal minimum singular value for arbitrary sender offsets. These are magnitudes in a selected real-coordinate metric, not cryptographic measures of signed-head divergence. The harness's achievability construction realizes an unrestricted edge perturbation; it does not show that a fixed sender can attain the same minimum.

Effective resistance maps where a one-edge fault is hard to see in edge-only data. Adding independent visible detours can lower the effective resistance of an edge and increase its one-edge sensitivity. Direct comparison remains the equal-information detection baseline, while scoring, localization, and audit cost need a common benchmark.

7 Sharp robustness under bounded error

An exact zero/nonzero residual is not an operational alarm when legitimate edge measurements have error. Fix one coordinate, one visible graph K_c, its incidence map B_c, and its cycle-space projector Π_c. Assume both the honest and faulty hypotheses permit *every* projected error vector of norm at most ϵ, with unrestricted vertex potentials under each hypothesis. This is a deterministic worst-case error contract, not a noise distribution or a claim about signed-head

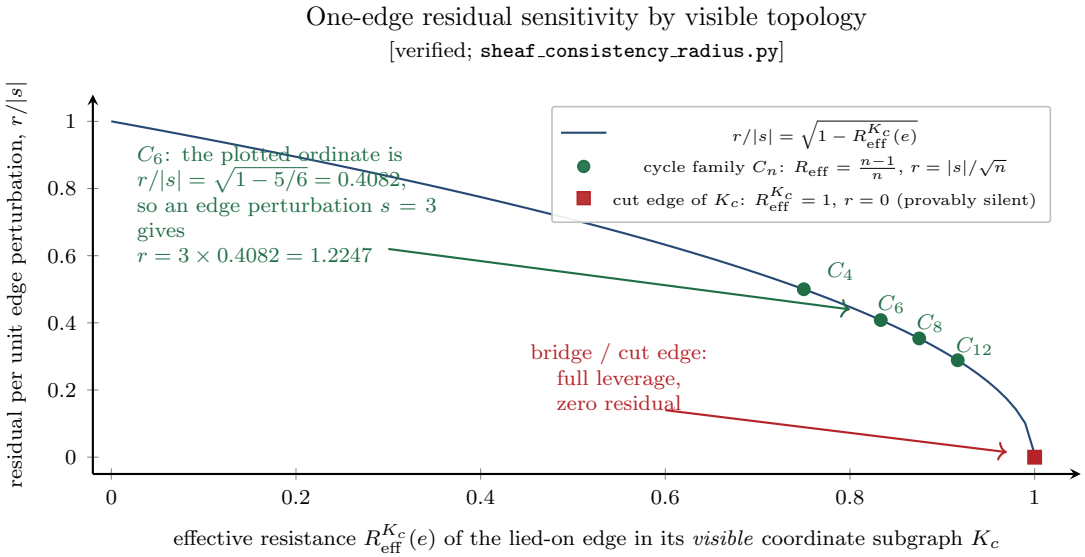


Figure 3: One-edge residual sensitivity is set by the visible topology and chosen Euclidean feature metric. The closed form plots residual *per unit edge perturbation*, $r/|s| = \sqrt{1 - R_{\text{eff}}^{K_c}(e)}$, against the effective resistance of the lied-on edge in its *visible* coordinate subgraph K_c — compared-plus-relayed links only, so severing a link on the edge's cycle moves the point right, toward silence. The cycle family C_n sits at $R_{\text{eff}}^{K_c} = \frac{n-1}{n}$; the marked C_6 point is the plotted ordinate 0.4082, which an absolute lie of $s = 3$ scales to the harness's $r = 1.2247$. The cut-edge corner $R_{\text{eff}}^{K_c} = 1$ is provable silence [verified; closed form exact on C_4–C_{12} and 200 random graphs, `sheaf_consistency_radius.py`].

uncertainty. The graph, feature map, coordinate scale, and snapshot are held fixed.

Theorem 3 (sharp bounded-error separation). Let e be a visible coordinate edge, let $s \neq 0$, and define

$$\mathcal{H}_0 = \{B_c x + \eta : \|\Pi_c \eta\| \leq \epsilon\}, \qquad \mathcal{H}_1(e, s) = \{B_c x + s\mathbf{e}_e + \eta : \|\Pi_c \eta\| \leq \epsilon\}.$$

These sets of possible edge observations are disjoint if and only if

$$|s|\sqrt{1 - R_{\text{eff}}^{K_c}(e)} > 2\epsilon. \tag{1}$$

When (1) holds, the rule $r(g) = \|\Pi_c g\| > \epsilon$ has zero worst-case false alarms and misses for this model. At equality the two classes share an observation. For an unknown fault edge in a specified nonempty visible set T_c with $|s| \geq s_{\min} > 0$, a uniform zero-error guarantee exists exactly when $s_{\min} \min_{e \in T_c} \sqrt{1 - R_{\text{eff}}^{K_c}(e)} > 2\epsilon$; a bridge makes the minimum zero.

Proof. Modulo $\operatorname{im} B_c$, the two classes are closed Euclidean ϵ-balls centered at 0 and $v = \Pi_c(s\mathbf{e}_e)$, respectively. Their center distance is the left side of (1), by Theorem 1. Balls of radius ϵ are disjoint exactly when that distance exceeds 2ϵ. Otherwise $v/2$ belongs to both classes, using errors $v/2$ and $-v/2$; unrestricted potentials absorb the gradient components. Under (1), honest observations have $r \leq \epsilon$ and faulty observations have $r \geq \|v\| - \epsilon > \epsilon$. Taking the minimum over allowed edges proves the uniform statement. □

The topological blind-spot dimension and the noise margin combine for *all* split views by one sender. Fix a sender q with at least two visible c-edges. Let U_q^c be the line of uniform offsets

on those edges, and define

$$\gamma_q^c = \min_{\operatorname{dist}(o,U_q^c)=1} \|\Pi_c A_q^c o\|.$$

Uniform offsets are coboundaries, so this is the smallest singular value of $\Pi_c A_q^c$ restricted to the orthogonal complement of U_q^c. The minimum is zero exactly when $b_q^c > 0$. If an adversary may choose any offset with $\operatorname{dist}(o, U_q^c) \geq \tau > 0$, the honest and one-sender-fault observation classes are disjoint for *every* such offset if and only if

$$b_q^c = 0 \quad \text{and} \quad \tau\gamma_q^c > 2\epsilon. \tag{2}$$

Indeed, when $b_q^c > 0$, an invisible nonuniform kernel offset can be scaled beyond τ. Otherwise the smallest projected signal is exactly $\tau\gamma_q^c$, attained by a least singular vector, and the ball argument above is necessary and sufficient. On a visible C_n at a degree-two sender, $b_q^c = 0$ and $\gamma_q^c = \sqrt{2/n}$; on a path's internal sender, $b_q^c > 0$ and $\gamma_q^c = 0$. For C_6 with a one-edge fault of size 3, the exact error budget is $\epsilon < 3/(2\sqrt{6}) \approx 0.6124$. For a degree-two sender on the same ring with nonuniform offset norm at least 1, the all-split-view budget is $\epsilon < 1/(2\sqrt{3}) \approx 0.2887$.

These are zero-error boundaries under the stipulated complete error balls. Merely knowing an upper bound on error makes (1) and (2) sufficient, but their necessity can fail for restricted or asymmetric errors. A directional test may be better under a probability model; no statistical optimality or deployed false-alarm rate follows. If authenticated endpoint claims are available, their comparison remains the equal-information detection baseline and requires its own error model.

8 The value of one more receipt

An auditor with a fixed edge-only receipt budget can choose which missing comparison to request next. A new row may close a visible cycle, but the mere presence of a cycle does not determine its measured contribution. Fix one coordinate and snapshot, an oriented incidence matrix B for the receipts already held, and their observed differences g. Write $L = B^{\mathsf{T}}B$, choose any least-squares solution $\hat{x}$, and let $r^2 = \min_x \|g - Bx\|^2$. A candidate receipt has oriented row b with one $+1$ and one -1 and reports y in the same declared feature metric. The candidate is an independently retained row.

Theorem 4 (exact marginal receipt value). If the endpoints of b lie in the same connected component of the old visible graph, the residual after retaining the new receipt obeys

$$r_{\text{new}}^2 - r^2 = \frac{(y - b\hat{x})^2}{1 + bL^+b^{\mathsf{T}}} = \frac{(y - b\hat{x})^2}{1 + R_{\text{eff}}^K(u, v)}. \tag{3}$$

The innovation $y - b\hat{x}$ is independent of the gauge chosen for $\hat{x}$. Its square is zero exactly when the new row agrees with the old least-squares endpoint difference. If the endpoints lie in different old components, $r_{\text{new}}^2 = r^2$ for every y. This includes a first link between isolated vertices.

Proof. Old least-squares orthogonality gives $\|g - B(\hat{x} + z)\|^2 = r^2 + \|Bz\|^2$. In a connected component, the least energy $\|Bz\|^2$ subject to $bz = t$ is $t^2/(bL^+b^{\mathsf{T}})$, by the electrical variational principle. Minimizing $t^2/(bL^+b^{\mathsf{T}}) + (y - b\hat{x} - t)^2$ over t gives (3). If the endpoints are in different components, their independent additive gauges realize any bz at zero old energy, so the new row is fitted without changing the old minimum. □

On the path 0–1–2 with old differences $(0, 0)$, an additional 0–2 receipt of $y = 3$ has $R_{\text{eff}} = 2$ and raises r^2 by $9/3 = 3$. This is an exact *marginal inconsistency score*, not a probability of dishonesty. For a negative control, two parallel old receipts with readings 0 and 2 yield a fitted endpoint difference of 1. A third reading of 1 has zero marginal gain although it disagrees with

both old readings; a third reading of 0 has gain 2/3 although it agrees with one. A bridge has zero immediate gain, but it can connect components so that a later receipt closes a cycle.

For a predeclared finite library of failure hypotheses, apply (3) to the *difference* between the two hypotheses' old edge signatures and candidate-edge signatures. It gives their exact increase in squared projected separation. A candidate with zero innovation for a pair cannot separate that pair through this new row. A cost-aware audit may rank candidate rows by a declared worst-pair margin per query cost; the ranking needs predicted signatures and costs, and no universal optimality follows. Actual observed y is unknown before a query, so (3) alone is not an ex ante information-gain promise. If authenticated endpoint claims are retained, direct sender-claim comparison is still the equal-information baseline.

9 Related work and contribution

Cellular sheaves on graphs and their coboundary calculus follow Curry [3]; spectral sheaf methods follow Hansen–Ghrist [4]. HodgeRank already decomposes edge observations into a best-fit gradient and a cyclic residual [5]. Effective resistance as an incidence-row leverage score is also established [7]. The language of local consistency without a global assignment is related to Abramsky–Brandenburger contextuality [2]. Robinson [8] supplies the consistency-radius perspective. The projection, cycle-space identity, and effective-resistance law here are familiar graph linear algebra rather than a new cohomology invariant. The marginal receipt formula is likewise a rank-one least-squares identity in graph form. Its use here is to state a checkable acquisition rule and its disconnected-component exception for this evidence contract; it is not a claim of a new resistance or experimental-design theorem.

Robust HodgeRank studies noisy and contaminated edge comparisons with cyclic projections and statistical outlier detection [6]. The bounded-error separation above is an exact statement for the paper's specified edge-only evidence and adversarial Euclidean error balls; it does not claim a new general robust-ranking method.

PeerReview [1] and BFT forensics [10] motivate preserving authenticated statements that can identify a conflicting sender. When both endpoint claims are available to the analyst, a direct duplicate-claim check has that evidentiary advantage. The present real-coordinate model instead characterizes the consistency score of derived edge differences. An exact cycle-sum checker sees the same noiseless inconsistency as its least-squares residual; the residual adds a metric magnitude and a topology-dependent distribution of error. The contribution is an explicit information contract, the one-edge sensitivity formula under visible-edge deletion, the general single-sender kernel characterization, sharp bounded-error margins for one-edge and all one-sender faults, and a counterexample that prevents extending the one-edge criterion to arbitrary split views.

10 Limits of the certificate

- A zero residual certifies only the existence of a potential for visible encoded edge differences. The triangle-with-leaf example is a one-sender split view with zero residual. Separate sender claims may still expose it. A coalition can also arrange canceling edge offsets.
- A nonzero residual does not name a culprit. Different signed-report histories can yield the same edge-difference vector; retaining authenticated claims is necessary for direct attribution.
- A severed edge contributes no evidence. Deleting an edge can also break the only visible cycle supporting a one-edge residual. This is a statement about the specified

edge-data pipeline, not a claim that all evidence about that witness is absent.

- The numerical radius depends on the equality-preserving feature map and metric. With measurement noise η, the honest residual may be as large as $\|\Pi_K \eta\|$; operational thresholds need a noise model or calibration. A zero real-coordinate obstruction does not establish consistency in every richer coefficient system [9].

Reproducibility

The sampled arms use seed 20260816. `sheaf_mechanism_proof.py` checks cycle and bridge witnesses; `sheaf_harness_v2.py` checks visibility, direct claims, bounded-error margins, and marginal receipt value; `sheaf_consistency_radius.py` checks resistance and radius calculations. The harness rejects any residual-only case against an equal-information claim checker. Its unrestricted completion construction does not solve the fixed-sender minimum; timings and localization are synthetic, and no signed-receipt deployment is tested. The triangle-with-leaf and componentwise-kernel arguments are independent hand checks.

References

[1] A. Haeberlen, P. Kouznetsov, and P. Druschel. PeerReview: practical accountability for distributed systems. In *Proc. 21st ACM Symposium on Operating Systems Principles (SOSP)*, pp. 175–188, 2007.

[2] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.

[3] J. M. Curry. *Sheaves, Cosheaves and Applications*. PhD thesis, University of Pennsylvania, 2014.

[4] J. Hansen and R. Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019.

[5] X. Jiang, L.-H. Lim, Y. Yao, and Y. Ye. Statistical ranking and combinatorial Hodge theory. *Mathematical Programming*, 127:203–244, 2011.

[6] Q. Xu, J. Xiong, X. Cao, Q. Huang, and Y. Yao. Evaluating visual properties via robust HodgeRank. arXiv:1408.3467, revised 2021.

[7] P. Drineas and M. W. Mahoney. Effective resistances, statistical leverage, and applications to linear equation solving. arXiv:1005.3097, 2010.

[8] M. Robinson. Sheaves are the canonical data structure for sensor integration. *Information Fusion*, 36:208–224, 2017.

[9] G. Carù. On the cohomology of contextuality. In *Proc. 13th International Conference on Quantum Physics and Logic (QPL 2016)*, EPTCS 236, pp. 21–39, 2017.

[10] P. Sheng, G. Wang, K. Nayak, S. Kannan, and P. Viswanath. BFT protocol forensics. In *Proc. ACM Conference on Computer and Communications Security (CCS)*, pp. 1722–1743, 2021.

[11] D. A. Spielman and S.-H. Teng. Nearly-linear time algorithms for graph partitioning, graph sparsification, and solving linear systems. In *Proc. 36th ACM Symposium on Theory of Computing (STOC)*, pp. 81–90, 2004.